PROJECT DOCUMENTATION

1. Title: **E-Commerce Sales and Customer Analysis**

2. Objective: The objective of this project is to perform **Exploratory Data Analysis (EDA)** on an e-commerce dataset to understand order patterns, sales performance, customer behavior, product performance, payment preferences, order statuses, and relationships between numerical variables.

The analysis aims to:

- Understand the structure and quality of the dataset.
- Identify patterns and trends in orders and sales.
- Analyze product-wise performance.
- Analyze payment method preferences.
- Examine order status distribution.
- Study customer purchasing behavior.
- Detect outliers in numerical variables.
- Analyze correlations between numerical variables.
- Generate meaningful business insights from the data.

3. Dataset Description: The dataset contains **1,200 e-commerce orders** with **14 columns**.

Dataset Dimensions

- **Rows:** 1,200
- **Columns:** 14
- **Date Range:** January 1, 2023 – June 30, 2025

Columns

Column	Description
OrderID	Unique identifier for each order
Date	Date on which the order was placed
CustomerID	Identifier of the customer
Product	Product purchased
Quantity	Number of units purchased
UnitPrice	Price of one unit

ShippingAddress	Customer shipping address
PaymentMethod	Payment method used
OrderStatus	Current status of the order
TrackingNumber	Shipment tracking identifier
ItemsInCart	Number of items in the customer's cart
CouponCode	Coupon applied to the order
ReferralSource	Source through which the customer arrived
TotalPrice	Total value of the order

4. Tools and Technologies Used

- **Python**
- **Google Colab**
- **Pandas** – Data manipulation and analysis
- **NumPy** – Numerical operations
- **Matplotlib** – Data visualization
- **Seaborn** – Statistical visualization
- **Jupyter/Google Colab Notebook** – Analysis environment

5. Data Understanding: The dataset was first examined to understand its structure, dimensions, column names, and data types.

Dataset Shape: (1200, 14)

Data Types :-

- Object columns → identifiers and categorical variables
- Integer columns → Quantity, ItemsInCart
- Float columns → UnitPrice, TotalPrice
- Datetime column → Date

The Date column was confirmed to have the datetime64[ns] data type.

6. Data Quality Analysis: Several checks were performed to verify the quality and consistency of the dataset.

Results:-

- **Missing values:** 0
- **Duplicate rows:** 0
- **Duplicate OrderIDs:** 0
- **Duplicate TrackingNumbers:** 0
- **Unique OrderIDs:** 1,200
- **Unique TrackingNumbers:** 1,200
- **Missing CustomerIDs:** 0

Text columns were also checked for leading/trailing spaces.

The following columns had **0 values containing unwanted leading or trailing spaces**:

- Product
- PaymentMethod
- OrderStatus
- CouponCode
- ReferralSource

Therefore, the dataset was found to be **clean and consistent**, requiring no major data-cleaning operations.

7. Descriptive Statistics: The major numerical variables were analyzed using descriptive statistics.

Variable	Count	Mean	Median
Quantity	1200	2.95	3.00
UnitPrice	1200	₹356.41	₹364.21
ItemsInCart	1200	5.49	5.00
TotalPrice	1200	₹1,053.97	₹823.62

Interpretation

- Customers purchased an average of approximately **3 units per order**.
- The average unit price was approximately **₹356.41**.
- The average number of items in a cart was approximately **5.49**.
- The average order value was approximately **₹1,053.97**.

- The median order value was lower than the mean, indicating that some higher-value orders influence the average.

8. Outlier Analysis: The **Interquartile Range (IQR)** method was used to detect potential outliers.

Variable	Outlier Count
Quantity	0
UnitPrice	0
ItemsInCart	0
TotalPrice	8

Interpretation

- Quantity, UnitPrice, and ItemsInCart had **no detected outliers** using the IQR method.
- TotalPrice contains **8 potential outliers**.
- These high-value orders were investigated individually rather than automatically removed.

The highest order value was **₹3,456.40**.

Because these values represent legitimate high-value orders in the dataset, they were **retained** for analysis.

9. Categorical Analysis: The distribution of major categorical variables was analyzed.

Product:-

Product	Orders
Printer	181
Tablet	179
Chair	178
Laptop	173
Desk	170
Monitor	163
Phone	156

Printer was the most frequently ordered product with **181 orders**.

Payment Method:-

Payment Method	Orders
Online	258
Cash	246
Credit Card	234
Debit Card	232
Gift Card	230

Online was the most frequently used payment method with **258 orders**.

Order Status:-

Status	Orders
Cancelled	250
Returned	247
Pending	237
Shipped	235
Delivered	231

Cancelled was the most common order status with **250 orders**.

Coupon Code:-

Coupon Code	Orders
FREESHIP	313
Unknown	309
WINTER15	292
SAVE10	286

FREESHIP was the most common coupon code.

Referral Source:-

Referral Source	Orders
Instagram	259
Email	250
Google	241
Facebook	228
Referral	222

Instagram was the most common referral source.

10. Time-Based Analysis:-

The order dates range from: **January 1, 2023 to June 30, 2025.**

Monthly order volumes were analyzed to identify temporal patterns.

The monthly order count fluctuated between approximately **27 and 53 orders**.

The highest monthly order count was: **June 2024 → 53 orders**

The lowest was: **January 2025 → 27 orders**

Year-wise Orders

Year	Orders
2023	510
2024	459
2025*	231

2025 contains only January–June data. Therefore, the 2025 total should **not be directly compared with the full-year totals**.

11. Sales Analysis:-

Year-wise Total Order Value:-

Year	Total Order Value
2023	₹552,643.24
2024	₹480,235.87
2025*	₹231,882.85

The highest yearly order value was recorded in **2023**.

The average order value by year was:

Year	Average Order Value
2023	₹1,083.61
2024	₹1,046.27
2025*	₹1,003.82

This indicates a gradual decrease in average order value across the available periods, although 2025 represents only six months.

12. Product Performance Analysis: Products were compared using order count, total sales, and average order value.

Product	Orders	Total Sales	Avg. Order Value
Chair	178	₹195,620.11	₹1,098.99
Printer	181	₹195,612.61	₹1,080.73
Laptop	173	₹192,126.56	**₹1,110.56**
Tablet	179	₹186,568.95	₹1,042.28
Monitor	163	₹175,651.41	₹1,077.62
Desk	170	₹167,459.93	₹985.06
Phone	156	₹151,722.39	₹972.58

Key Findings:-

- **Chair** generated the highest total sales: **₹195,620.11**.
- **Printer** had the highest order count: **181**.
- **Laptop** had the highest average order value: **₹1,110.56**.
- **Phone** had the lowest total sales: **₹151,722.39**.

13. Quantity Analysis: Total quantity sold by product was analyzed.

Product	Total Quantity
Chair	562
Printer	542
Laptop	535
Desk	508
Tablet	497
Monitor	480
Phone	411

Finding: Chair had the highest total quantity sold with **562 units**, while **Phone** had the lowest with **411 units**.

14. Payment Method Analysis:-

Payment Method	Orders	Total Sales	Avg. Order Value
Credit Card	234	₹263,847.63	₹1,127.55
Online	258	₹262,442.94	₹1,017.22
Cash	246	₹259,786.29	₹1,056.04
Gift Card	230	₹246,323.92	₹1,070.97
Debit Card	232	₹232,361.18	₹1,001.56

Finding:-

- **Online** had the highest number of orders.
- **Credit Card** generated the highest total order value.
- Credit Card also had the highest average order value.

This demonstrates that the most frequently used payment method is not necessarily the method associated with the highest-value orders.

15. Order Status Analysis:-

Order Status	Orders	Total Order Value	Avg. Order Value
Cancelled	250	₹276,396.21	₹1,105.58
Pending	237	₹256,328.15	₹1,081.55
Shipped	235	₹246,159.58	₹1,047.49
Returned	247	₹243,277.70	₹984.93
Delivered	231	₹242,600.32	₹1,050.22

Finding: Cancelled orders were the most common status.However, the TotalPrice associated with cancelled or returned orders should **not automatically be interpreted as realized revenue**, because the dataset does not provide sufficient information to determine actual revenue after cancellations and returns.

16. Correlation Analysis: Correlation analysis was performed on the numerical variables.

Relationship	Correlation

UnitPrice – TotalPrice	**0.717**
Quantity – TotalPrice	**0.615**
ItemsInCart – TotalPrice	**0.393**
Quantity – ItemsInCart	**0.650**
Quantity – UnitPrice	**0.015**
UnitPrice – ItemsInCart	**0.001**

Interpretation

- UnitPrice has the strongest positive correlation with TotalPrice.
- Quantity also has a strong positive relationship with TotalPrice.
- ItemsInCart has a moderate positive relationship with TotalPrice.
- Quantity and UnitPrice have almost no correlation.

This is logical because:

TotalPrice = Quantity × UnitPrice

17. Customer Analysis:-

The dataset contains: **1,189 unique customers across 1,200 orders.**

No customer placed more than **2 orders**.

The highest-value customer was: **C38840 → ₹5,723.23 across 2 orders**

with an average order value of: **₹2,861.62**

Finding: The dataset is dominated by customers with a single order, indicating relatively limited repeat purchasing within the available data.

18. Overall Key Findings:-

The major findings from the analysis are:

1. The dataset contains **1,200 clean e-commerce orders** with no missing values or duplicate records.
2. **Printer** was the most frequently ordered product.
3. **Chair** generated the highest total sales and had the highest quantity sold.
4. **Laptop** had the highest average order value among products.
5. **Online** was the most commonly used payment method.
6. **Credit Card** generated the highest total order value.

7. **Cancelled** was the most common order status.
8. **Instagram** was the most common referral source.
9. **FREESHIP** was the most common coupon code.
10. UnitPrice had the strongest correlation with TotalPrice.
11. The dataset contained **8 potential outliers in TotalPrice**, which were retained as legitimate high-value orders.
12. Customer activity was dominated by single-order customers.

19. Conclusion:-

The exploratory data analysis provided a comprehensive understanding of the e-commerce dataset from product, customer, payment, order-status, temporal, and numerical perspectives.

The analysis showed that **Chair and Printer were among the strongest products**, while Credit Card transactions were associated with the highest total order value and average order value. Online payments were the most frequently used, while cancelled orders represented the largest order-status category.

The correlation analysis showed that **UnitPrice and Quantity are the primary numerical factors associated with TotalPrice**.